Tree of Thoughts: Deliberate Problem Solving with Large Language Models

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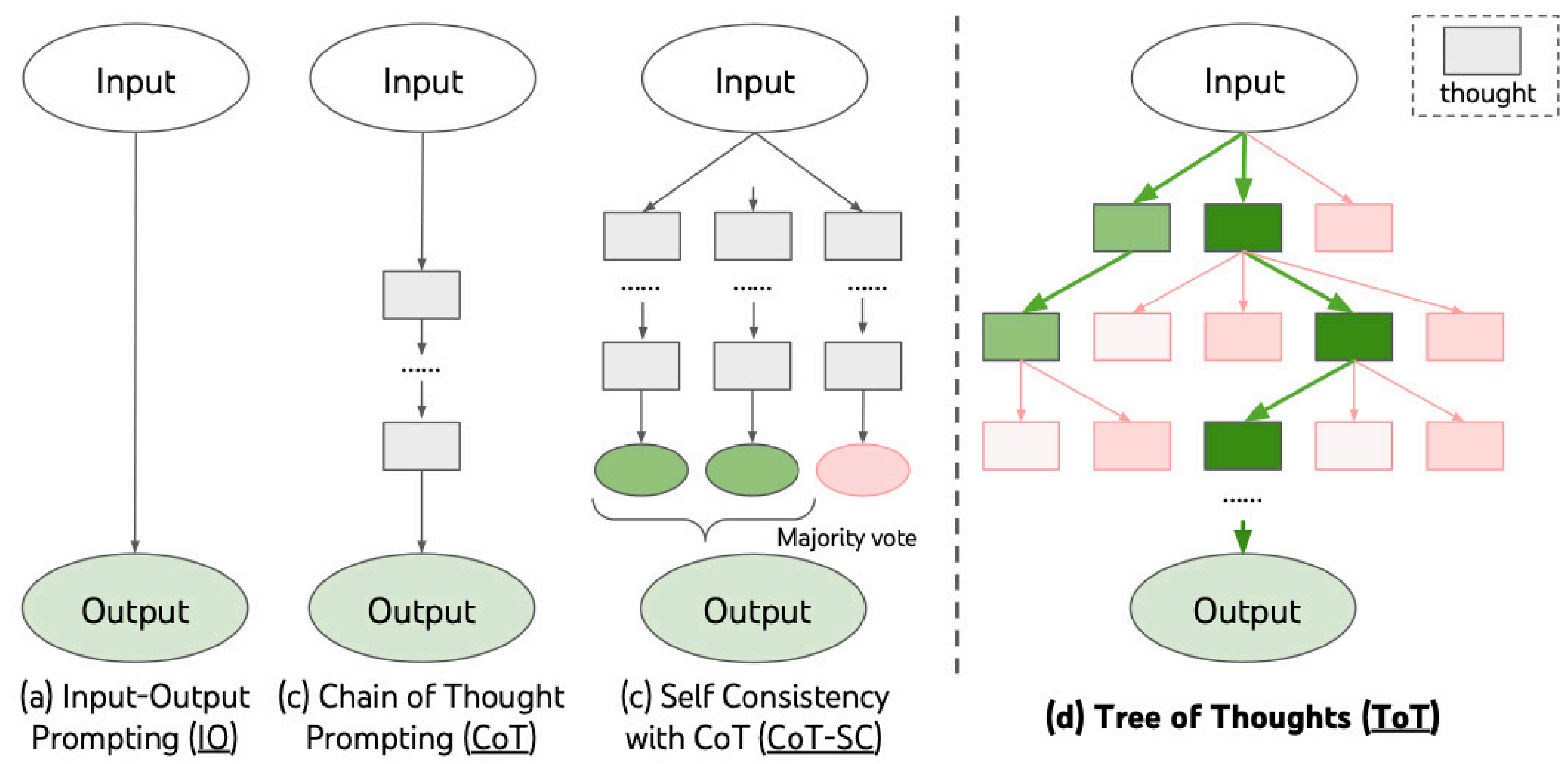
Background

- LLMs are confined to token-level, left-to-right decision making processes
- They can fall short in tasks that require exploration/strategic lookahead
- Yao et al. (2023) proposed novel framework - **Tree of Thoughts (ToT)** - to address this challenge
- Experiments on 3 tasks - Game of 24, Creative Writing, Crosswords - show ToT significantly improves performance

Goal: Reproduce 3 tasks in paper + Implement new task 4x4 Sudoku

Motivation

How to approach general problem solving using LLMs?



Results

Game of 24

Method	Success (Ours)	Success (Paper)
IO Prompt	6.8%	7.3%
CoT Prompt	3.1%	4.0%
ToT (b=5)	67%	74%

4x4 Sudoku

Method	Success
IO Prompt	9.2%
CoT Prompt	12%
ToT (b=5)	85%

Methodology

Model:

Qwen3.6

- 28B parameters

Dataset:

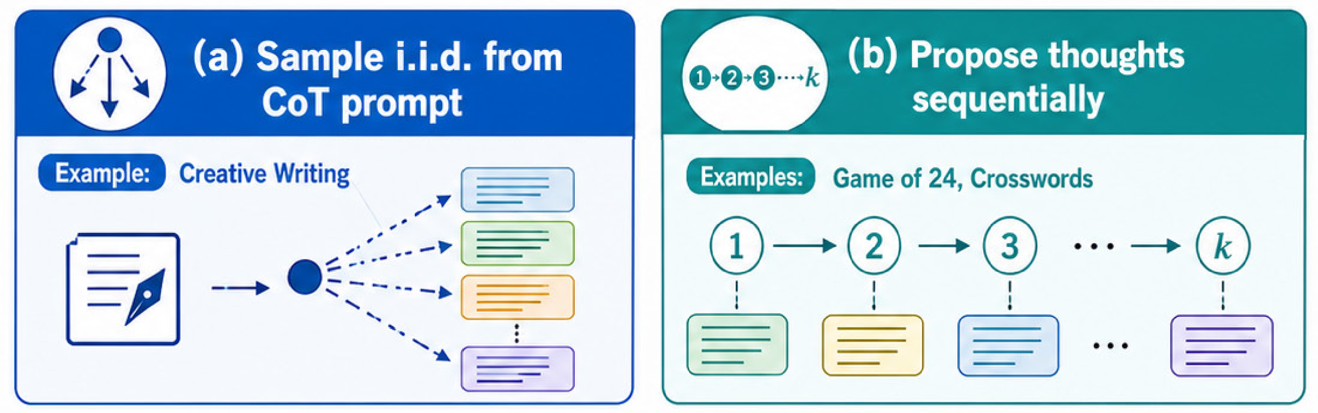
- Puzzles for all 3 tasks available in paper repo
- 1 million unique 4x4 Sudoku puzzles downloaded from public repo by Black-Phoenix

Step 1 - Thought Decomposition

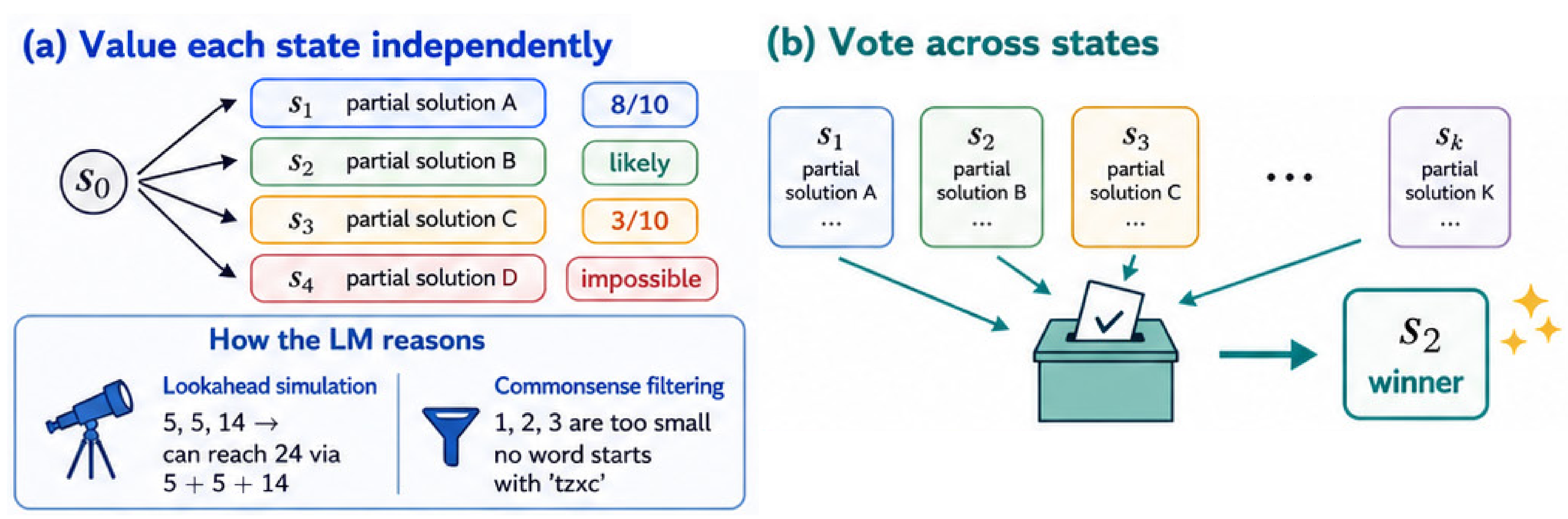
	3	1	2
2			4
3			1
	4		

$(2 + 8) = 10$

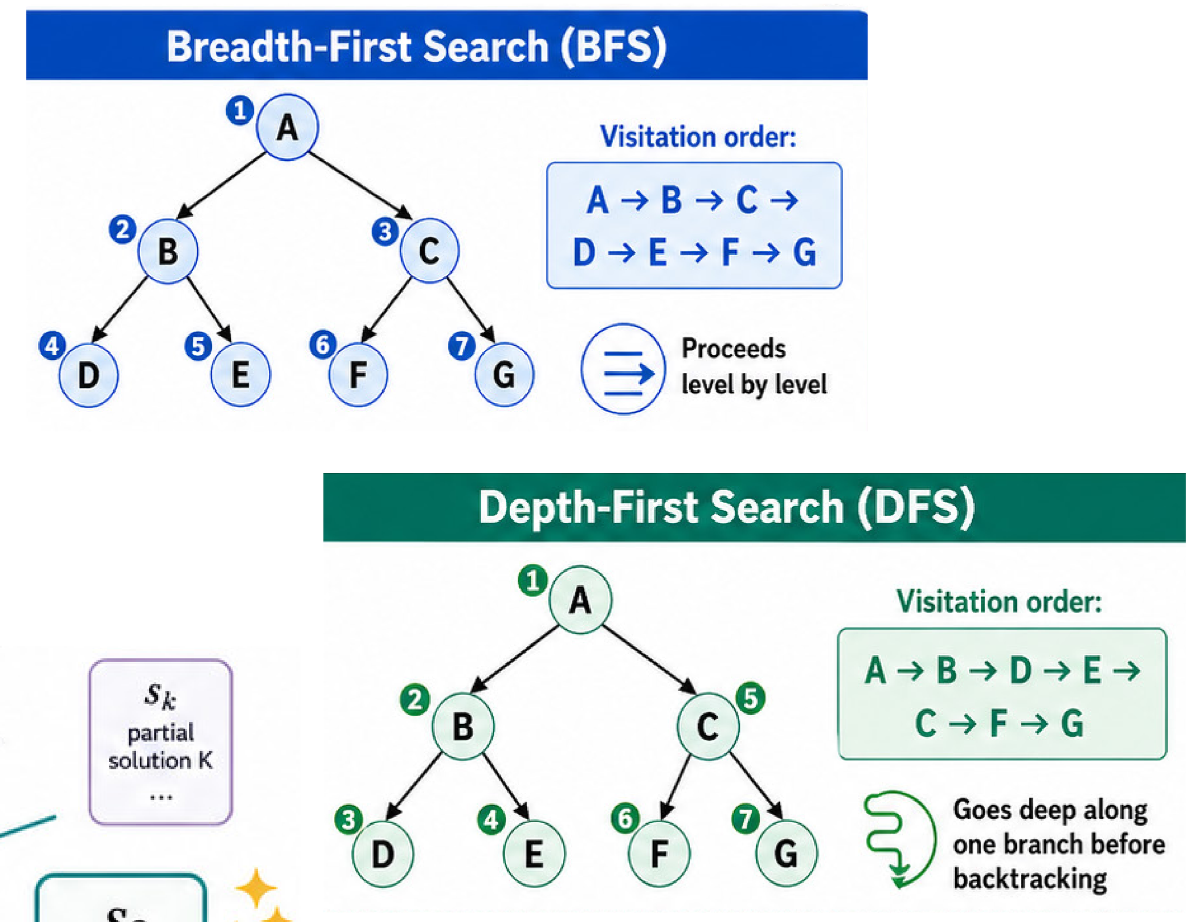
Step 2 - Thought Generator



Step 3 - State Evaluator



Step 4 - Search Algorithm



Conclusion & Future Work

- Our results for Game of 24 match closely with paper's result
- Performance for 4x4 Sudoku seems nominally better because test sample size was smaller
- Next step: reproduce results for Creative Writing and Crosswords

References

Yao, Shunyu, et al. "Tree of Thoughts: Deliberate Problem Solving with Large Language Models." ArXiv.org, 17 May 2023, arxiv.org/abs/2305.10601.

Black-Phoenix. "GitHub - Black-Phoenix/4x4-Sudoku-Dataset: 1 Million 4x4 Sudoku Boards." GitHub, 2025, github.com/Black-Phoenix/4x4-Sudoku-Dataset. Accessed 27 Apr. 2026.